Chapter 6

Emerging Risk Measures

The previous two chapters developed two complementary article-level representations of the news corpus. First, thematic relevance scores ($S_{i,k}$) quantify the extent to which each article is associated with each of the five predefined risk categories. Second, document embeddings (E_i) provide a general semantic representation of the information contained in each article.

These representations provide the inputs required to introduce the temporal dimension of the analysis. The objective of this chapter is to transform article-level textual information into quantitative measures capable of describing different manifestations of financial risk emergence. A central premise of the empirical framework is that emergence is not a single directly observable phenomenon. A risk may begin to emerge because new information appears that is substantially different from previously observed news. The meaning and mechanisms associated with an existing risk category may change over time. A risk may receive an increasing share of media attention. Finally, attention may increase sufficiently rapidly and materially to constitute an exceptional attention episode rather than an ordinary fluctuation in coverage. The empirical framework therefore distinguishes four complementary dimensions:

1. **Novelty**, measuring whether the semantic information contained in an article is unusual relative to recent historical news;
2. **Semantic Drift**, measuring whether the semantic representation associated with a risk category changes over time;
3. **Coverage**, measuring the relative amount of news attention devoted to a risk category;
4. **Burst Detection**, identifying periods in which thematic attention experiences an unusually sharp and material increase relative to its own historical dynamics.

These measures capture distinct dimensions of the information environment and should not be interpreted as alternative estimates of the same latent quantity. Novelty is initially defined at the article level. Semantic drift and coverage are constructed at the theme–month level. Burst detection is subsequently applied to the monthly coverage series. The resulting analytical sequence can be summarized as

$$\boxed{\text{Thematic Scores + Document Embeddings + Time} \longrightarrow \text{Emerging Risk Measures}}. \tag{6.1}$$

6.1 Novelty

The first dimension of emergence considered in this study is the semantic novelty. The goal is to see if an article contains information that was already covered in earlier articles. If not, the article brings new content from a semantic point of view. Because every article is represented by a normalized document embedding, novelty can be defined as a local distance in the semantic space. If an article is similar to earlier articles, most of its content is already present in previous news. Its novelty is therefore expected to be low. In contrast, if an article differs from earlier articles, its content is less common in previous news. Its novelty is therefore expected to be higher. The measure therefore relies on the semantic neighbourhood of each article rather than on its absolute location in the embedding space.

6.1.1 Historical Semantic Neighbourhood

For each article i, a historical candidate set is defined as

$$\mathcal{H}_i(W) = \left\{ j : t_i - W \leq t_j < t_i \right\}, \tag{6.2}$$

where t_i denotes the publication date of article i and W is the historical look-back window. Within this historical information set, the K articles with the highest cosine similarity to the current article are identified. Let

$$NN_K(i; W) \tag{6.3}$$

denote the set of these K nearest historical semantic neighbours. The general novelty measure is

$$N_i(K, W) = 1 - \frac{1}{K} \sum_{j \in NN_K(i;W)} \cos(E_i, E_j). \tag{6.4}$$

Since all document embeddings are L_2-normalized,

$$\cos(E_i, E_j) = E_i^\top E_j. \tag{6.5}$$

The measure can therefore be interpreted as one minus the average semantic similarity between the current article and its closest historical neighbours.
A high average similarity produces a low novelty score and a lower average similarity produces a higher novelty score.

Novelty is a semantic-distance measure rather than a probability. A value of $N_i = 0.20$, for example, should not be interpreted as implying that an article is "20% new".

6.1.2 Choice of the Number of Neighbours and Historical Window

The novelty measure depends on two main parameters: the number of nearest neighbours K and the historical window W. Both parameters were selected through a calibration exercise on the news corpus. The aim was not to maximize or minimize novelty. Instead, the calibration looked for a balance between historical coverage, semantic locality, and score stability.
Three historical windows were tested:

$$W \in \{30, 90, 365\} \text{ days}, \tag{6.6}$$

together with five values for the number of neighbours:

$$K \in \{1, 5, 10, 20, 50\}. \tag{6.7}$$

The resulting grid contained 15 alternative specifications. The pilot used 1,080 target articles, producing 16,200 article-specification observations.

For each specification, the candidate set included only articles within the selected historical window:

$$d_i - W \leq d_j < d_i. \tag{6.8}$$

The strict inequality means that only information published before the target article can enter its reference set. Articles from the same calendar date are excluded.
This restriction is important because reliable intraday publication times are not available for all articles. A same-day article could have been published after the target article. Including it could therefore introduce look-ahead contamination into the historical benchmark.

The parameter W controls how much past information is available when searching for semantic neighbours. As expected, longer windows produced much larger candidate sets. With $W = 30$, the median candidate set contained about 16,090 articles. It increased to 47,766 for $W = 90$ and 204,664 for $W = 365$.
The window length also affected the level of measured novelty. With $K = 10$, mean novelty was 0.330 for the 30-day window. It fell to 0.297 for 90 days and 0.265 for 365 days.
This pattern has a simple explanation. A longer window gives the model more chances to find a relevant historical precedent. With a shorter window, related information may fall outside the reference period and appear new.
However, using the entire historical corpus would create a different issue. Very old articles could affect whether current information is considered novel. The measure would then become less connected to the recent information environment.
For these reasons, $W = 365$ was selected as the baseline. It provides one year of historical information while keeping the reference period clearly defined. This choice is part of the methodology and is not imposed by the embedding model.
The 365-day window also creates a boundary condition at the start of the sample. The corpus begins in 2014. Articles from that year do not have a complete year of previous observations.
Therefore, 2014 is used as a warm-up period for the novelty measure. The main analytical period runs from 2015 to 2023. This prevents missing pre-sample information from being interpreted as unusually high semantic novelty.

The parameter K plays a different role. It determines how many historical articles form the local semantic benchmark.
With $K = 1$, novelty depends only on the closest historical article. The score can therefore be sensitive to a near-duplicate or one unusually similar story.

A large K creates the opposite issue. It gradually includes less similar articles, making the benchmark broader but less local in semantic terms.

Table 6.1 summarizes the main calibration results. Panel A compares the three historical windows while keeping $K = 10$. Panel B compares alternative values of K within the 365-day window.

Table 6.1: Novelty calibration results

Panel A: Historical window ($K = 10$)

W (days)	K	Mean novelty	Spearman correlation
30	10	0.330	0.884
90	10	0.297	0.945
365	10	0.265	1.000

Panel B: Number of neighbours ($W = 365$)

W (days)	K	Mean novelty	Spearman correlation
365	1	0.210	0.880
365	5	0.245	0.988
365	10	0.265	1.000
365	20	0.287	0.990
365	50	0.320	0.953

Note: Spearman correlations are computed against the final specification, $K = 10$ and $W = 365$ days.

Panel A shows that mean novelty falls as the historical window becomes longer. A larger reference set makes it easier to find relevant historical precedents. Article rankings remain fairly stable across the three windows.
Panel B shows what happens when the number of neighbours changes. Values close to $K = 10$ produce very similar article rankings. The Spearman correlation is 0.988 for $K = 5$ and 0.990 for $K = 20$. It falls to 0.953 when K increases to 50.

Overall, the results support a broad historical window combined with a relatively local semantic neighbourhood. The final specification therefore uses

$$\boxed{K = 10} \tag{6.9}$$

and

$$\boxed{W = 365 \text{ days}}. \tag{6.10}$$

The choice of $K = 10$ balances stability and semantic locality. The score does not depend on one historical article, while the neighbourhood remains relatively small.
The 365-day window limits false novelty caused by a short historical memory. At the same time, the comparison remains linked to a recent information environment.

Importantly, financial-market outcomes were not used to select K or W. The calibration only considered the behaviour and stability of the novelty measure. Both parameters were therefore fixed before the later market analysis.

Given these choices, the final article-level novelty measure is

$$N_i = 1 - \frac{1}{10} \sum_{j \in NN_{10}(i;365)} \cos(E_i, E_j), \tag{6.11}$$

where $NN_{10}(i; 365)$ contains the ten most similar historical articles from the previous 365 days, subject to $d_j < d_i$.

6.1.3 Computational Implementation

The corpus contains more than two million document embeddings, each with 384 dimensions. Comparing every article with all eligible historical articles would therefore require a very large number of computations.

To make the process feasible, nearest-neighbour search was implemented using FAISS/HNSW. This method allows similar historical articles to be identified efficiently across the full corpus.

The computational method should be kept separate from the definition of novelty itself. Equation (6.11) defines how the novelty score is calculated. FAISS/HNSW is instead used to find the relevant historical neighbours efficiently.

The resulting novelty scores are stored at article level in

`risk_emergence.news_novelty`.

Novelty is therefore first computed for individual articles. It remains distinct from the monthly theme-level indicators introduced in the following sections.

6.2 Semantic Drift

Novelty evaluates individual articles. Semantic drift captures a different aspect of emergence: changes in the semantic content associated with a risk category.

A risk label may remain unchanged while the events, technologies, institutions, or consequences linked to that risk evolve. Coverage alone cannot capture this change. More articles indicate greater attention, but they do not show whether the content of that attention is also changing.

Semantic drift addresses this issue through a monthly semantic representation of each risk category. This representation is compared with its recent historical state. The procedure requires three main choices. These concern the representation of each theme, the selection of reliable observations, and the construction of the historical benchmark.

6.2.1 Combining Thematic Scores and Document Embeddings

The theme-specific semantic representation combines two article-level outputs developed in the previous chapters. The document embedding E_i represents the semantic content of article i. The thematic score $s_{i,k}$ measures the strength of its association with risk category k.

For semantic drift, the thematic signal is the frozen weakly supervised continuous score:

$$s_{i,k} = S^{WS}_{i,k}. \tag{6.12}$$

Only articles with positive thematic evidence are included:

$$s_{i,k} > 0. \tag{6.13}$$

For each theme k and month t, these articles are combined into a score-weighted semantic centroid:

$$\tilde{c}_{k,t} = \frac{\sum_{i \in (k,t): s_{i,k} > 0} s_{i,k} E_i}{\sum_{i \in (k,t): s_{i,k} > 0} s_{i,k}}. \tag{6.14}$$

The thematic score determines the contribution of each article. Articles with stronger thematic evidence therefore receive greater weight.
The resulting vector is then L_2-normalized:

$$c_{k,t} = \frac{\tilde{c}_{k,t}}{\|\tilde{c}_{k,t}\|_2}. \tag{6.15}$$

The normalized centroid $c_{k,t}$ represents the semantic position of risk category k in month t.

6.2.2 Choice of the Thematic Signal

The thematic signal was selected by comparing the empirical properties of the available scoring methods. The comparison focused mainly on the weakly supervised and supervised approaches.
On the independent final test, the weak continuous method achieved

$$\text{ROC-AUC}_{WS} = 0.785135, \tag{6.16}$$

$$\text{AP}_{WS} = 0.583126, \tag{6.17}$$

whereas the supervised V3 model achieved

$$\text{ROC-AUC}_{SUP} = 0.742859, \tag{6.18}$$

$$\text{AP}_{SUP} = 0.112327. \tag{6.19}$$

The weak score showed stronger ranking performance, especially in terms of Average Precision. It was therefore selected as the thematic signal for semantic drift.
However, the weak signal is highly sparse. About 95.8% of its observations in the final test were equal to zero. For this reason, the reliability of monthly semantic representations could not be assumed. A separate selection and validation procedure was needed.

6.2.3 Selection Strategy and Reliability Validation

Several selection rules were tested before defining the final monthly centroids.
The first pilot used a common absolute threshold:

$$S_{i,k,m} \geq 0.5. \tag{6.20}$$

The rule was applied to all three thematic-scoring methods. It generated 1,736 of the 1,800 potential method–theme–month centroids.

However, the three methods produce scores through different mechanisms and have different empirical distributions. A value of 0.5 does not necessarily indicate the same level of thematic evidence across methods. The common absolute threshold was therefore rejected.

A second pilot used a relative selection rule. Within each method–theme–month group, the top 10% of observations were selected by thematic score. This avoided imposing the same numerical threshold on different score distributions.

However, this approach created a problem for the weakly supervised signal. In 54 weakly supervised theme–month groups, all observations in the top decile had a score of zero. In these cases,

$$\sum_i s_{i,k} = 0, \tag{6.21}$$

so a score-weighted centroid could not be constructed.

The problem was also conceptual. If all scores are zero, a fixed percentile selects articles through arbitrary ties. Yet the model provides no positive thematic evidence for those articles. The top-decile approach was therefore rejected.

The final strategy uses the positive support of the weak continuous score:

$$s_{i,k} > 0. \tag{6.22}$$

This rule includes only articles for which the model provides positive thematic evidence. It also avoids an additional absolute or percentile threshold.

The amount of positive evidence varies across risk categories. Cyber, climate, and supply-chain risk show positive thematic evidence throughout the sample. Pandemic risk is very sparse before 2020, with 54 months containing no positive observations. AI-related risk also has less positive support than the more established categories.

A monthly centroid should therefore not be considered reliable simply because one positive article exists. For this reason, a minimum-support requirement was evaluated empirically.

A preliminary analysis showed that consecutive monthly centroids became more similar as thematic support increased. However, this comparison cannot separate estimation noise from genuine semantic change. A split-half reliability procedure was therefore used.

For each theme–month group, positive observations were divided deterministically into two subsets. An independent score-weighted centroid was then constructed for each subset:

$$c_{k,t}^{(A)} \qquad \text{and} \qquad c_{k,t}^{(B)}. \tag{6.23}$$

Their cosine similarity was calculated as

$$R_{k,t} = \cos\left(c_{k,t}^{(A)}, c_{k,t}^{(B)}\right). \tag{6.24}$$

Both centroids describe the same risk category in the same month. Low similarity therefore suggests an unstable semantic representation rather than genuine temporal drift.

The validation used ten deterministic splits. For groups with at least ten positive observations, overall reliability was

$$\text{mean}(R_{k,t}) = 0.966312, \tag{6.25}$$

$$\text{median}(R_{k,t}) = 0.985228, \tag{6.26}$$

$$p05(R_{k,t}) = 0.864616. \tag{6.27}$$

Reliability increased clearly with thematic support, as shown in Table 6.2.

Table 6.2: Split-half centroid reliability by thematic support

Positive articles	Mean	5th percentile	Minimum
10–19	0.774	0.678	0.644
20–29	0.847	0.799	0.781
30–49	0.888	0.830	0.813
50–99	0.928	0.900	0.878
100–299	0.972	0.955	0.916
300–999	0.988	0.981	0.978
≥ 1000	0.996	0.993	0.992

These results were then used to choose the minimum level of thematic support. With

$$n_{\text{positive}} \geq 10, \tag{6.28}$$

520 of the 600 potential theme–month centroids would be retained. However, minimum split-half reliability would be only about 0.644.

Higher thresholds improve reliability but reduce temporal coverage. This trade-off is especially relevant for sparse categories such as AI-related risk.

At

$$n_{\text{positive}} \geq 50, \tag{6.29}$$

461 of the 600 potential centroids are retained. The corresponding reliability statistics are

$$\text{mean}(R) = 0.980656, \tag{6.30}$$

$$p05(R) = 0.932592, \tag{6.31}$$

$$\min(R) = 0.878467. \tag{6.32}$$

A threshold of 100 would further improve average reliability. However, it would remove a substantial number of observations for AI-related risk. The final specification therefore uses

$$\boxed{n_{\text{positive}} \geq 50}. \tag{6.33}$$

This threshold provides a balance between semantic reliability and temporal coverage.

An effective-sample-size diagnostic was also computed:

$$N_{\text{eff}} = \frac{(\sum_i s_i)^2}{\sum_i s_i^2}. \tag{6.34}$$

This statistic helps identify groups dominated by a small number of articles with high weights. The diagnostic did not support an additional production threshold. The final specification therefore retained the simpler $n_{\text{positive}} \geq 50$ rule.

6.2.4 Historical Reference and Semantic Drift

After constructing a reliable current centroid, semantic drift requires a historical benchmark. Three alternatives were considered.
The first option was a month-on-month comparison:

$$D_{k,t}^{MoM} = 1 - \cos(c_{k,t}, c_{k,t-1}). \tag{6.35}$$

This measure is easy to interpret but has two limitations. It cannot be computed when the previous month lacks a reliable centroid. It can also be sensitive to short-term variation.
The second option was a fixed historical baseline:

$$D_{k,t}^{Fixed} = 1 - \cos(c_{k,t}, c_{k,t_0}). \tag{6.36}$$

This approach depends strongly on the choice of t_0. As time passes, it also measures increasing displacement from a single historical state.
The final specification uses a rolling historical reference. For each current month t, the reference can use reliable centroids from the previous 12 calendar months:

$$\mathcal{H}_{k,t} = \left\{ c_{k,\tau} : 1 \leq t - \tau \leq 12 \right\}. \tag{6.37}$$

Only reliable historical centroids are included. Missing months are not imputed. Current and future information never enters the historical reference.
Let

$$h_{k,t} = |\mathcal{H}_{k,t}| \tag{6.38}$$

denote the number of available historical centroids. At least three are required:

$$h_{k,t} \geq 3. \tag{6.39}$$

The historical monthly centroids receive equal weight. Their unnormalized mean is

$$\tilde{r}_{k,t} = \frac{1}{h_{k,t}} \sum_{c_{k,\tau} \in \mathcal{H}_{k,t}} c_{k,\tau}. \tag{6.40}$$

The historical reference is then L_2-normalized:

$$r_{k,t} = \frac{\tilde{r}_{k,t}}{\|\tilde{r}_{k,t}\|_2}. \tag{6.41}$$

The two weighting stages therefore serve different purposes. Articles within a theme–month are weighted by their thematic scores. Historical monthly centroids instead receive equal weight. As a result, months with more articles do not automatically dominate the historical benchmark.
The final semantic-drift measure is

$$\boxed{D_{k,t} = 1 - \cos(c_{k,t}, r_{k,t})}. \tag{6.42}$$

Higher values indicate greater semantic displacement from the recent historical representation of the same risk category.

Missing observations are kept separate from zero semantic drift. A reliable current centroid is not constructed when a theme–month contains fewer than 50 positive articles. Semantic drift is also unavailable when fewer than three reliable historical centroids exist in the previous 12 months.

In either case,

$$D_{k,t} = \text{NA}. \tag{6.43}$$

A missing value indicates that the available information is insufficient to estimate semantic change reliably. This differs from $D_{k,t} = 0$. A zero value means that a valid current representation coincides with its historical benchmark.

6.2.5 Production Results

The final production procedure processed 408,027 positive article–theme observations. Positive thematic mass was present in 546 of the 600 possible theme–month combinations.

After the minimum-support rule was applied, 461 reliable monthly centroids remained. The rolling historical-reference requirements then produced 443 valid semantic-drift observations.

Table 6.3 reports the distribution of the final measure.

Table 6.3: Distribution of the final semantic-drift measure

Statistic	Semantic Drift
Minimum	0.006654
Mean	0.029467
Median	0.023402
95th percentile	0.067083
Maximum	0.154270

The final semantic-drift values were stored in the monthly theme-level indicator structure.

6.3 Coverage

Semantic drift measures changes in the content associated with a risk category. It does not measure how much attention that category receives. Coverage provides this complementary dimension by measuring the presence of each risk in the monthly news flow.

6.3.1 Definition and Choice of the Coverage Measure

For theme k and month t, let N_t denote the total number of valid articles published during the month. Two alternative coverage measures were considered.

The first uses the thematic scores directly:

$$C_{k,t}^{weighted} = \frac{\sum_{i \in t} s_{i,k}}{N_t}. \tag{6.44}$$

This measure combines two components: the number of related articles and the strength of their thematic scores.

The second measure uses positive thematic evidence:

$$C_{k,t}^{binary} = \frac{\#\{i \in t : s_{i,k} > 0\}}{N_t}. \tag{6.45}$$

It has a direct interpretation as the share of valid monthly articles with positive evidence for risk theme k.

The two definitions were compared before selecting the production specification. Table 6.4 reports their monthly correlations for each risk category.

Table 6.4: Correlation between binary and weighted monthly coverage

Risk Theme	Correlation
Cyber Risk	0.976537
Climate Risk	0.998625
Pandemic Risk	0.999777
Supply Chain Risk	0.998478
AI-Related Risk	0.996814

The correlations are high for all five risk categories. The two definitions therefore produce very similar temporal patterns.

The binary-positive measure was selected for final production. Its main advantage is its simple interpretation as a share of the monthly news flow. It also avoids giving additional importance to differences in the magnitude of thematic scores.

Let

$$N_{k,t} = \#\{i \in t : s_{i,k} > 0\} \tag{6.46}$$

denote the number of articles with positive evidence for theme k in month t. Final monthly coverage is then defined as

$$\boxed{x_{k,t} = C_{k,t} = \frac{N_{k,t}}{N_t}}. \tag{6.47}$$

The measure ranges from zero to one. Higher values indicate that a larger share of the monthly news flow is associated with the risk category.

The normalized coverage series $x_{k,t}$ is then used as the input to the burst-detection procedure developed in Section 6.4.

6.3.2 Coverage Results

The sample contains 120 months, from January 2014 to December 2023. With five risk categories, the resulting panel contains 600 theme–month observations.

Table 6.5 reports positive thematic observations and the main coverage statistics for each category.

Table 6.5: Monthly coverage by risk theme

Theme	Total Positive	Min. Monthly	Max. Monthly	Mean Coverage	Max. Coverage
Cyber Risk	38,664	110	715	0.017405	0.032366
Climate Risk	120,017	477	1,948	0.059039	0.128928
Pandemic Risk	195,075	0	11,974	0.108873	0.753556
Supply Chain Risk	47,787	143	1,169	0.024822	0.076560
AI-Related Risk	6,484	4	214	0.003529	0.018551

Coverage is defined for all 600 theme–month observations. The results also show large differences in the presence of the five risks in the news.

Pandemic risk has the highest mean coverage, at 0.108873, and reaches a maximum of 0.753556. Its monthly count ranges from zero to 11,974 positive articles. This wide range reflects the strong concentration of pandemic-related news in part of the sample.

Climate risk has the second-highest mean coverage, at 0.059039. Supply-chain and cyber risk have lower average values, at 0.024822 and 0.017405, respectively. AI-related risk has the lowest mean coverage, at 0.003529.

A zero value is a valid coverage observation. If no article in month t has positive evidence for theme k, then

$$C_{k,t} = 0. \tag{6.48}$$

This treatment differs from semantic drift. When thematic support is insufficient for a reliable semantic representation, the corresponding drift observation is missing rather than zero:

$$D_{k,t} = \text{NA}. \tag{6.49}$$

The distinction is important. Zero coverage indicates an observed absence of positive thematic evidence in the monthly news flow. Missing semantic drift instead indicates that semantic change cannot be estimated reliably.

6.4 Burst Detection

Coverage measures the amount of news attention linked to each risk category. However, it does not separate normal variation from unusually sharp increases in attention.

Burst detection adds this second dimension. Its purpose is to identify exceptional increases relative to the recent history of each risk. The method also separates the onset of an attention shock from its persistence over time.

6.4.1 Historical Benchmark

Burst detection is applied to the normalized monthly coverage series $x_{k,t}$, defined in Equation (6.47). Normalized coverage is used instead of raw article counts because the total volume of news changes over time.

For each theme, the historical benchmark is based on the previous 12 months. The rolling mean is

$$\mu_{k,t}^{(12)} = \frac{1}{12} \sum_{j=1}^{12} x_{k,t-j}, \tag{6.50}$$

while the corresponding historical standard deviation is

$$\sigma_{k,t}^{(12)} = \mathrm{sd}\left(x_k, t-12, \ldots, x_k, t-1\right). \tag{6.51}$$

The current observation $x_{k,t}$ is excluded from both statistics. The benchmark is therefore strictly backward-looking. Information from month t cannot affect the historical distribution used to assess that same month.

A standardized coverage shock is then calculated as

$$Z_{k,t}^{(12)} = \frac{x_{k,t}\mu^{(12)}k,t}{\sigma^{(12)}k,t}. \tag{6.52}$$

This statistic measures how unusual current coverage is relative to the recent variability of the same risk category.

6.4.2 Burst Onset, Persistence, and Intensity

A high value of $Z_{k,t}^{(12)}$ is not enough on its own to identify a burst onset. When historical coverage is very low, even a small absolute increase can generate a large standardized deviation.

For this reason, the onset rule also includes a historical materiality condition. Let

$$Q_{0.95,k,t}^{\mathrm{hist}} = Q_{0.95}\left(x_{k,1}, \ldots, x_{k,t-1}\right) \tag{6.53}$$

denote the empirical 95th percentile of all coverage observations available for theme k before month t. This percentile uses the full available history of the theme rather than only the 12-month rolling window.

If the theme is not already in an active burst, a new episode begins when

$$\boxed{Z_{k,t}^{(12)} > 3 \qquad \text{and} \qquad x_{k,t} > Q_{0.95,k,t}^{\mathrm{hist}}}. \tag{6.54}$$

The two conditions have different roles. The z-score identifies an unusual increase relative to recent dynamics. The historical percentile requires coverage to also be high relative to the broader history of that theme. This filter reduces signals caused by large standardized changes from very low coverage levels.

Burst detection is implemented as a state process. Once an episode starts, the onset conditions are not applied again in every following month. An active burst continues while

$$x_{k,t} > \mu_{k,t}^{(12)}. \tag{6.55}$$

The episode ends in the first month for which

$$x_{k,t} \leq \mu_{k,t}^{(12)}. \tag{6.56}$$

Any new onset signal observed during an active episode is absorbed into the existing burst. It is

therefore not counted as a separate episode.
The resulting burst-state variable is

$$B_{k,t} = \begin{cases} 1 & \text{if theme } k \text{ is in an active burst in month } t, \\ 0 & \text{otherwise.} \end{cases} \tag{6.57}$$

The binary state shows whether a burst is active, but it does not measure its size. A separate intensity variable is therefore defined as

$$I_{k,t} = \begin{cases} x_{k,t} - \mu^{(12)} k, t & \text{if } Bk, t = 1, \\ 0 & \text{if } B_{k,t} = 0. \end{cases} \tag{6.58}$$

The onset statistic and the intensity measure therefore capture two different aspects of a burst. The z-score detects unusual changes relative to recent variability. Burst intensity instead measures excess attention in units of normalized coverage.
This distinction matters when historical variability is very small. In these cases, standardized shocks can become very large even when the absolute change in coverage is limited.

6.4.3 Specification and Calibration

The final specification was selected through a set of diagnostic exercises before storing the burst indicators in the final dataset.
Several alternatives were considered. They included different historical windows, robust dispersion measures, count-based and binomial formulations, alternative materiality filters, and different state rules. Frozen historical benchmarks and alternative measures of burst intensity were also examined.
A 12-month backward-looking window was retained for the historical benchmark. It gives a local comparison while still covering a full annual period.
For the onset condition, standardized thresholds of 2.5, 3.0, and 3.5 were compared. The final specification uses

$$\boxed{Z_{k,t}^{(12)} > 3}. \tag{6.59}$$

The lower threshold generated additional weak episodes. In contrast, the higher threshold removed or delayed several major increases in thematic attention. A threshold of three standard deviations gave a more balanced result.
The all-prior 95th-percentile filter was retained as the materiality condition. During calibration, it removed standardized spikes linked to very low coverage levels. At the same time, it preserved the main attention episodes identified by the rolling (z)-score.
Alternative persistence rules were also tested. One option was to freeze the historical benchmark when a burst started. However, this rule produced overly persistent episodes when thematic coverage moved to a more lasting level.
The final specification instead updates the 12-month benchmark every month. This allows the burst state to adjust gradually when the underlying attention environment changes.
Alternative intensity measures were considered as well. One option was relative excess coverage,

$$\frac{x_{k,t}}{\mu_{k,t}^{(12)}} - 1, \tag{6.60}$$

which becomes unstable when historical coverage is close to zero. A bounded alternative was

$$\frac{x_{k,t} - \mu^{(12)} k, t}{xk, t}. \tag{6.61}$$

This measure avoids the same instability, but it compresses large attention episodes. For this reason, the absolute-excess measure in Equation (6.58) was retained.

The final specification separates detection from measurement. Onset depends on a standardized shock together with a historical materiality filter. Persistence requires coverage to remain above its moving historical mean. Intensity then measures the absolute excess above that benchmark.

6.4.4 Warm-Up Period

The rolling benchmark requires 12 prior monthly observations. Therefore, the first 12 months available for each theme form a warm-up period.

During these months,

$$B_{k,t} = I_{k,t} = \text{NA}, \qquad t = 1, \ldots, 12. \tag{6.62}$$

A missing burst state is different from a state equal to zero. A value of $B_{k,t} = 0$ means that the month was evaluated and no active burst was found. A missing value instead means that the required historical information was not yet available.

6.4.5 Production Results

The procedure was applied to the five risk categories over the 2014–2023 sample. The complete panel contains 600 theme–month observations.

For each theme, the first 12 months form the warm-up period. This gives 60 non-evaluable observations and leaves 540 evaluable theme–month observations. Of these, 110 are classified as active burst months and 430 as inactive months. Overall, the procedure identifies 19 distinct burst episodes.

Table 6.6 reports the distribution of the final burst indicators across the five risk categories.

Table 6.6: Burst-detection results by risk theme

Theme	Active Burst Months	Episodes	Maximum Burst Intensity
Cyber Risk	15	3	0.01310225
Climate Risk	22	4	0.05103656
Pandemic Risk	12	1	0.68381492
Supply Chain Risk	35	5	0.04025854
AI-Related Risk	26	6	0.01165945
Total	110	19	–

The results show that burst frequency and burst magnitude capture different features of news attention. Supply-chain risk records the most active burst months, with 35 months across five episodes. AI-related

risk instead has the largest number of distinct episodes, with six episodes covering 26 months.
Pandemic risk follows a different pattern. It has only one burst episode, which lasts 12 months, but reaches a maximum intensity of 0.683815. This value is substantially higher than the maximum intensity recorded for any other theme. It reflects the exceptional rise in pandemic-related news coverage during 2020.
Cyber risk records 15 active burst months across three episodes. These occur in December 2018–November 2019, January 2020, and February–March 2022. They provide the burst component of the cyber-risk indicators used in the later empirical analysis.
The final values were stored in the monthly theme-level indicator dataset as `burst_state` and `burst_intensity`. The database write was checked against the final pre-commit output. All 600 theme–month observations matched, with no differences in either burst state or burst intensity.

6.5 Joint Interpretation of the Emerging Risk Measures

The four measures developed in this chapter capture distinct but complementary dimensions of financial risk emergence. They describe changes in information content, thematic meaning, news prevalence, and attention dynamics.

Novelty operates at the article level. It captures whether an article contains semantic information that differs from the recent historical news environment. A high novelty score therefore signals the arrival of relatively unusual information. It does not imply that the corresponding risk already receives broad media attention.

Semantic drift operates at the theme–month level. It measures whether the current semantic representation of a risk category differs from its recent historical representation. A risk can therefore display semantic drift even when its label remains unchanged. The measure captures changes in the events, mechanisms, or consequences associated with that category.

Coverage provides a different perspective. It measures the share of monthly news with positive evidence for a given risk category. Higher coverage indicates greater thematic prevalence in the news flow. However, it does not show whether the underlying content is new or whether the increase in attention is historically unusual.

Burst detection adds this final distinction. It identifies unusually sharp and material increases in coverage relative to the theme's own historical dynamics. Burst intensity then measures the amount of excess coverage during an active episode. A risk can therefore have high coverage without being in a burst if that level of attention is not exceptional relative to its recent history.

The conceptual distinction between the four measures is summarized in Table 6.7.

Table 6.7: Interpretation of the emerging-risk measures

Measure	Main Question
Novelty	Is the information in the article semantically new relative to recent historical news?
Semantic Drift	Is the meaning associated with the risk category changing relative to its recent historical representation?
Coverage	How much of the current news flow is associated with the risk category?
Burst Detection	Is thematic attention increasing in an unusually sharp and material way relative to its own history?

The measures should therefore be interpreted as complementary rather than as substitutes. Different combinations can describe different forms of risk emergence.
For example, high coverage combined with low semantic drift describes a risk receiving substantial attention while its semantic content remains relatively stable. High semantic drift with low coverage instead suggests that the meaning of a risk is changing while the category remains relatively uncommon in the overall news flow.
Novelty provides another distinction. Semantically unusual articles may appear before a risk becomes widely represented in the media. In this case, novelty can increase without an immediate rise in aggregate coverage. Conversely, high coverage can result from repeated discussion of already familiar information and therefore coexist with relatively low novelty.
Coverage and burst detection also need to be interpreted separately. Persistently high coverage does not necessarily represent an emerging attention shock. Burst detection requires the increase to be exceptional relative to the historical behaviour of the same theme.

Taken together, the four measures provide a multidimensional representation of financial risk emergence. They separate changes in information content, thematic meaning, prevalence, and attention dynamics. The resulting news-based indicators provide the basis for the subsequent empirical analysis.